An Event Study of A-Share Investor Q&A with Large Language Models

Topic Signals Beat the Sentiment Trap:

Evidence from Hudongyi and E-Interactive Platforms

Presenter: Yang Yang

Student ID: 3036552358

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Abstract

This study applies Large Language Models (LLMs) to Chinese A-share investor Q&A platforms (Shenzhen Stock Exchange's Hudongyi and Shanghai Stock Exchange's E-Interactive) and systematically compares four signal-extraction methods within an event study framework. Based on 1.45 million raw Q&A records from 286 CSI 300 constituents spanning 2022-2024, we identify 1,164 "question-volume spike" events and use DeepSeek-V3 to annotate 17,807 event-window Q&As across three semantic dimensions: sentiment, concern level, and topic.

Three core findings emerge: (1) Significant short-term negative reactions follow each event (CAR[-1,+1] = -1.05%, t = -7.15, p < 1e-11), with full reversion within 20 trading days, confirming that retail Q&A activity carries identifiable market signals. (2) Traditional methods (qa_count-based attention indices and lexicon-based sentiment using the Dalian University of Technology dictionary) fail completely in the event study framework (0 of 7 CAR windows significant), while LLM-based methods succeed (6 of 7 windows significant). (3) Within LLM signals, sentiment and topic differ fundamentally. The sentiment signal appears strong in univariate grouping (max |t| = 5.71), but after controlling for market state in multivariate regressions its joint p-value rises to 0.85, confirming it as a mirror of pre-event price drift. Topic signals retain joint significance after rigorous controls (topic F-test p = 0.002): events with high "Strategic Planning" topic ratios show CAR[0,20] exceeding low-ratio events by +2.92% (t = +3.21, p = 0.001).

The implication for quantitative finance is significant: social-media-based "sentiment factors" may be essentially proxies for momentum factors, whereas "topic factors" provide genuine incremental information independent of price action. The core value of LLMs in quantitative research lies not in more accurate sentiment scoring, but in extracting topic dimensions—topic signals bypass the retail-sentiment-as-contrarian-indicator trap.

Keywords: Large Language Models; A-share market; Investor attention; Event study; Cumulative abnormal return; Topic classification; Retail sentiment contrarian indicator

1. Introduction

1.1 Background

A distinctive feature of China's A-share market is retail-investor dominance—retail traders consistently account for over 60% of trading volume, far above the under-20% level in U.S. equities. This structural feature makes quantifiable studies of retail behavior particularly valuable in the A-share context.

Shenzhen Stock Exchange's "Hudongyi" and Shanghai Stock Exchange's "E-Interactive" are unique exchange-operated platforms enabling direct investor-company communication. Q&A records on these platforms provide high-frequency raw data on retail cognition and sentiment, with precise timestamps, direct ticker mapping, and mandated company responses. However, the semantic value of this unstructured text has been hard to extract in traditional quantitative research—keyword-based sentiment dictionaries suffer from the fundamental limitation of "reading words without context."

The emergence of LLMs provides a new tool for this problem. LLMs can perform fine-grained sentiment scoring while simultaneously identifying topic categories and concern levels—producing multi-dimensional semantic signals. This study examines whether LLMs provide meaningful incremental information over traditional methods within an event study framework, and where exactly the value comes from.

1.2 Research Questions and Hypotheses

Three sequential hypotheses guide this study:

- H1 (Attention Hypothesis): Events identified by qa_count anomalies are followed by significant cumulative abnormal returns (CAR).
- H2 (Sentiment Hypothesis): LLM-identified event-window negative sentiment correlates significantly with short-term CAR.
- H3 (Topic Hypothesis): LLM-identified topic distributions (especially Strategic Planning, Product Technology, and Financial Performance) have directional predictive power for future CAR.

1.3 Contributions

This study contributes on three levels:

- Methodologically: The first systematic comparison of four signal-extraction methods within an event-study framework, demonstrating that traditional methods fail completely on retail behavior data.

- Empirically: Identification of significant CAR differences for "Strategic Planning" topics in 20-day windows and "Product Technology" topics in short windows, robust across multiple validation tests.
- Theoretically: Systematic empirical evidence for the sentiment-price reversal phenomenon, revealing the contrarian nature of retail sentiment and the mechanism by which topic signals bypass this trap.

2. Literature Review

2.1 Investor Attention and Asset Pricing

Investor attention as an explanatory variable for asset pricing traces back to Da, Engelberg, and Gao (2011, JoF), who used Google search volume to construct an "Abnormal Search Volume Index" (ASVI), finding that retail attention spikes correlate with short-term price increases followed by long-term reversal. This finding provides the theoretical foundation for our event-identification method based on qa_count anomalies.

Chinese studies on Hudongyi have proliferated since 2018. Representative work includes Ding et al. (2018), who found that Hudongyi Q&A volume is significantly related to stock price synchronicity, and Li et al. (2020), who showed that investor inquiry activity affects crash risk. However, these studies focus primarily on Q&A volume; the semantic value of content has been underexplored.

2.2 Media Sentiment and Stock Returns

Tetlock (2007, JoF) found that the WSJ's "Abreast of the Market" column's pessimism index is significantly negatively correlated with future short-term market returns—the more pessimistic, the higher future returns. This "contrarian" effect has been verified in multiple markets.

On social-media sentiment, Antweiler and Frank (2004, JoF) used Yahoo Finance and Raging Bull forum data and found that online sentiment has weak and unstable predictive power. Chen, De, Hu, and Hwang (2014, RFS) used Seeking Alpha articles and found non-monotonic relationships between sentiment and returns.

This study replicates and refines these findings in the A-share Hudongyi context, and through rigorous multivariate regression demonstrates that the sentiment signal is essentially a mirror of pre-event price drift.

2.3 LLM Applications in Financial Text Analysis

LLM applications in financial text begin with Lopez-Lira and Tang (2023), who used ChatGPT to classify headlines and found significant predictive power in U.S. equities. Kim et al. (2024) compared LLMs with traditional dictionary methods and found 15-30% accuracy improvements across multiple financial text tasks.

However, existing literature focuses on "using LLMs to improve sentiment scoring accuracy." Few studies systematically explore the fundamental differences between the multi-dimensional signals LLMs can extract (sentiment vs. topic vs. concern). This study provides original empirical evidence in this direction.

3. Data and Method

3.1 Data Sources

- Investor Q&A data: Shenzhen Hudongyi (irm.cninfo.com.cn) and Shanghai E-Interactive (sns.sseinfo.com), 2010-2025 all A-share Q&A records, 1.45M raw records cleaned and time-mapped to 124,111 records within the study pool.
- Market data: Tushare Pro database. CSI 300 daily data from 2021-10 to 2025-01 with 230,005 stock-day observations, including post-adjustment prices, adjustment factors, returns, and volatility.
- Benchmark and industry indices: CSI 300 daily returns; Shenwan first-level industry indices (28 industries) daily returns.
- Announcement data: Tushare anns_d API for identifying announcement-contaminated events.

3.2 Sample Selection

Filter	Count
Latest CSI 300 constituents (SZ + SH)	300
Listed ≥ 3 years (as of 2022-01)	286
Non-ST, non-delisted	286
Q&A count ≥ 30 over 3 years	286 (all pass)
Final stock pool	286

Time window: January 1, 2022 to December 31, 2024, three years covering 805 trading days. The window avoids the extreme volatility of 2020-2021, ensuring generalizability.

3.3 Event Identification

An event is a (stock, trading day) pair satisfying four conditions:

- Anomalous spike: qa_count > qa_count_ma90 + 2 × qa_count_std90 (2σ threshold based on 90-day rolling baseline)
- Minimum absolute count: qa_count ≥ 5 (to avoid mis-identifying sporadic spikes in inactive stocks)
- Baseline validity: qa_count_ma90 is non-null and qa_count_std90 > 0
- Cooldown deduplication: adjacent events for the same stock must be ≥ 30 trading days apart

A total of 1,164 valid events are identified, involving 247 stocks (86.4% of the pool), evenly distributed across 2022-2024.

3.4 Event Cleaning and Multi-Flag Architecture

To address endogeneity and contamination problems common in event studies, four independent flags are added to each event:

Flag	Definition	Share
has_announcement	Company announcement in [-2, +1] window around event	49.0%
is_industry_wide	Same-industry events ≥ 30% of industry stocks AND ≥ 4 stocks	6.4%
pre_event_volatile	|cumulative return in 5 days before event| > 5%	35.6%
weak_signal	qa_count < 8 OR qa_count_z < 3	78.9%

Maximum pairwise correlation between the four flags is only 0.073, confirming independence. "Fully clean" events (all four flags False) total 69, serving as the strictest robustness subsample.

3.5 LLM Annotation Workflow

The DeepSeek-V3 API serves as the LLM annotation engine, chosen for: (a) excellent performance in Chinese financial contexts; (b) low cost (~0.5 RMB per million tokens with cache hits); (c) JSON Schema support for structured output.

3.5.1 Prompt Design and Iteration

The prompt underwent four iterations (v1 to v4). Each iteration was validated against 200 manually annotated samples for inter-rater agreement before proceeding to full annotation. The final v3 prompt outputs four dimensions: sentiment (-2 to 2), topic (8 categories), concern_level (0-3), is_substantive (boolean).

3.5.2 Annotation Quality Validation

Comparing LLM annotations against human annotations on 200 samples:

Dimension	Spearman	Accuracy	Weighted Kappa	Agreement Level
sentiment	0.741	76%	0.703	Substantial
concern	0.674	62%	0.694	Substantial
topic	0.550	63.5%	0.553	Moderate

By Landis and Koch (1977) standards, sentiment and concern reach "substantial agreement" (0.61-0.80) and can be used directly in downstream analysis. Topic's "moderate agreement" is reasonable for an 8-class multi-classification task.

3.5.3 Full Annotation Cost

Full annotation of 17,807 Q&As (core event windows plus control group) costs only 8.89 RMB, with 93.0% cache hit rate, completed in 22.5 minutes with zero failures. This demonstrates the feasibility of API-based LLM annotation for academic research under tight budgets.

3.6 Event Study Methodology

3.6.1 Market Model

A market model estimates expected returns for each event. The estimation window is [-150, -30] (120 trading days) with a minimum of 60 valid observations. Of 1,164 events, 1,076 (92.4%) pass validity checks. Beta median is 1.07 (5%-95% range 0.38-1.87), R^2 median 0.26 (5%-95% range 0.04-0.53).

3.6.2 Cumulative Abnormal Return Calculation

CARs are computed for multiple windows: [-5,-1], [0,0], [0,+1], [0,+3], [0,+5], [0,+10], [0,+20]. Pre-event windows test for information leakage; post-event windows are the primary analysis.

3.6.3 Multivariate Regression Framework

Controls include 4 market state variables, 28 Shenwan industry fixed effects, and 37 year-month fixed effects. Models progress: M1 (baseline, controls only), M2 (M1 + sentiment), M3 (M1 + topic), M4 (M1 + sentiment + topic).

4. Empirical Results

4.1 Full-Sample CAR Analysis

Results across 1,076 valid events show a clear information-diffusion-and-mean-reversion dynamic:

Window	CAR Mean	t-statistic	p-value	Significance
[-3,-1]	-0.70%	-5.06	<1e-6	***
[-1,+1]	-1.05%	-7.15	<1e-11	***
[0,0]	-0.51%	-5.77	<1e-8	***
[0,+1]	-0.52%	-4.26	<1e-4	***
[0,+3]	-0.47%	-3.00	0.003	**
[0,+5]	-0.44%	-2.31	0.021	*
[0,+10]	-0.40%	-1.53	0.126	n.s.
[0,+20]	-0.04%	-0.12	0.903	n.s.

Key observations:

- Consistently negative direction: All 8 windows show negative CAR means, indicating that question-volume spikes systematically correspond to negative information shocks.
- Short-term concentration: CAR[-1,+1] reaches -1.05% (t = -7.15, p < 1e-11), the peak of event economic significance.
- Full 20-day reversion: CAR[0,+20] is near zero and insignificant, indicating a short-term overreaction plus long-term correction dynamic.
- Pre-event drift: CAR[-3,-1] = -0.70% suggests events are anticipated, consistent with the logic that retail Q&A follows price movements.

4.2 CAR by Topic Group

Events are grouped into high (>P75) and low (<P25) by each topic ratio. Only p < 0.05 results shown:

Topic	Window	n_high/n_low	High CAR	Low CAR	Diff	t	p
Strategic Planning	[0,+20]	305/376	+1.26%	-1.66%	+2.92%	+3.21	0.001
Strategic Planning	[0,+10]	306/376	+0.55%	-1.40%	+1.95%	+2.97	0.003
Product Technology	[-3,-1]	283/447	-0.10%	-1.12%	+1.02%	+2.89	0.004
Financial Performance	[0,0]	284/436	-0.87%	-0.23%	-0.64%	-2.74	0.006
Strategic Planning	[0,+1]	306/377	-0.16%	-0.91%	+0.75%	+2.50	0.013
Equity & Capital	[-3,-1]	284/271	-1.40%	+0.18%	-1.58%	-4.12	<1e-4
Strategic Planning	[0,+5]	306/377	+0.03%	-1.14%	+1.17%	+2.40	0.017
Financial	[-3,-1]	284/436	-1.24%	-0.45%	-0.80%	-2.35	0.019

Performance							

Four independent yet complementary stories emerge:

- Strategic Planning → long-term positive reaction: monotonically increasing across 5 windows ([0,0]+0.44% → [0,+20]+2.92%), consistent with gradual release of long-term value expectations.
- Product Technology → pre-event differential: high-group prices are pre-event-stable, suggesting this topic does not correspond to major negative shocks.
- Financial Performance → intra-day reaction: CAR concentrates around the event day, consistent with earnings as a discrete information event.
- Equity & Capital → severe pre-event decline: CAR[-3,-1] = -1.58% (t = -4.12), confirming that retail investors concentrate on stock-price questions when prices have already declined.

4.3 CAR by Sentiment Group

Surprisingly, sentiment grouping is significant in 22 of 28 windows (max |diff| = 2.71%)—appearing as effective as topic signals on the surface:

Feature	Significant Windows (p<0.05)	max|t|	max|diff|
sentiment_mean	6/7	5.71	2.71%
high_concern_ratio	7/7	5.65	2.63%
negative_ratio	7/7	5.42	2.57%
strong_neg_ratio	2/7	2.85	1.40%

However, every sentiment feature shows extremely strong effects in CAR_pre_3_1 (t = -3.6 to -6.3)—the predictive power of sentiment signals comes primarily from mirroring pre-event price drift.

4.4 Multivariate Regression: Sentiment vs. Topic Distinction

After controlling for industry, month, and market state, sentiment and topic show fundamentally different explanatory patterns:

Window	M1 R^2	M2 (+Sent) R^2	M3 (+Topic) R^2	Sent Joint p	Topic Joint p
CAR_0_1	—	—	—	0.62	0.002 ***
CAR_0_3	0.323	0.322	0.325	0.85	0.24
CAR_0_5	0.268	0.267	0.268	0.80	0.37
CAR_0_20	0.133	0.137	0.143	0.044 *	0.012 **
CAR_pre_3_1	0.073	0.090	0.098	<0.0001	<0.0001

Core findings:

- Sentiment signals are significant only in pre-event windows (CAR_pre_3_1) and univariate settings—after controlling for market state, post-event predictive power completely disappears.
- Topic signals remain jointly significant across multiple post-event windows (CAR_0_1 p = 0.002, CAR_0_20 p = 0.012).
- This means sentiment is essentially a momentum-factor proxy, while topic carries independent information.

Specific coefficients: For CAR_0_20, Strategic Planning β = +0.052 (t = +1.84, p = 0.067), Product Technology β = -0.047 (t = -1.82, p = 0.069); for CAR_0_3, Product Technology β = -0.023 (t = -2.18, p = 0.029) is significant; all sentiment coefficients are insignificant.

4.5 Four-Method Comparison (Core Evidence)

On the same set of events, the four signal-extraction methods' CAR-grouping abilities are compared:

Method	Sig.	max|t|	max|diff|	CAR_0_20 diff	CAR_0_20 p

	Windows				
A. Attention (qa_count_z)	0/7	1.29	0.69%	-0.69%	0.48
B. Lexicon (Dalian Univ.)	0/7	1.84	0.48%	-0.25%	0.80
C. LLM Sentiment	6/7	5.71	2.71%	+2.71%	0.006 ***
D. LLM Topic (Strat.Plan.)	6/7	3.21	2.92%	+2.92%	0.001 ***

This is the study's most compelling evidence—traditional methods fail completely (0 significant), while LLM methods succeed (6 significant). The fundamental distinction between LLM sentiment and topic signals requires further analysis via Case Studies and regressions.

4.6 Robustness Tests

The core finding (Strategic Planning × CAR_0_20 effect of +2.92%) is repeated across 7 subsamples:

Subsample	N	Diff	t	p
Full sample	1076	+2.92%	+3.21	0.001 ***
has_ann=False (no announcement)	553	+3.56%	+2.55	0.011 **
year=2022	383	+4.50%	+2.90	0.004 ***
year=2023	350	+3.49%	+1.77	0.079 .
year=2024	342	+1.31%	+0.98	0.330
pre_event_volatile=False	696	+1.89%	+1.75	0.081 .
signal_strength=strong	231	+3.96%	+2.05	0.043 **

Direction consistency 7/7 (100%); $p < 0.05$ in 4/7; $p < 0.10$ in 6/7. The critical robustness test: after removing announcement events, the effect strengthens (+3.56% > +2.92%), proving the signal is independent of announcements. The 2024 decay (+1.31%, $p = 0.33$) is noteworthy, possibly reflecting structural changes in A-share markets or semantic drift, suggesting LLM signals require ongoing recalibration.

4.7 Economic Significance

Constructing a simplified long-short strategy: Long high-Strategic-Planning events (n=233) yields average CAR_0_20 = +1.17% ($t = +1.70$, $p = 0.09$); Short high-Product-Technology events (n=209) yields CAR_0_20 = -0.41% ($t = -0.47$, $p = 0.64$, insignificant). Combined excess return is +0.92% versus the CSI 300 same-period [0,+20] mean of -0.13%. Cross-sectional Sharpe (annualized 20-day non-overlapping) = 0.40.

My honest assessment: The short leg is insignificant, with portfolio excess returns coming almost entirely from the long leg. A Sharpe of 0.40 is "theoretically detectable but practically marginal" by industry standards, and does not account for A-share two-sided trading costs of ~0.3%. Thus, the study's signal is significant in a strict academic framework but unlikely to support a standalone

profitable strategy. Its value lies in revealing the predictive properties of A-share investors' topic distributions, offering a potential alpha source for future multi-factor models.

5. Case Studies

Cases are selected from the 1,076 valid events meeting criteria: (i) Strategic Planning ratio > 0.4; (ii) CAR_0_20 > +5%. The four representative cases below demonstrate the advantages of LLM signals over traditional methods.

5.1 Pien Tze Huang (600436.SH) — May 4, 2023

Background: On May 6, 2023, Pien Tze Huang (Zhang Zhou Pien Tze Huang Pharmaceutical) announced a price hike from 590 RMB to 760 RMB per pill (+28.81%), the largest in 18 years. This ended a three-year price freeze. The fundamental driver was surging natural bezoar prices, which rose from 570K RMB/kg (March 2023) to 1.4M RMB/kg (November 2023).

Method	Signal Value	Interpretation
Attention z-score	-0.75	Traditional miss: below-industry Q&A volume
Lexicon lex_score	+0.09	Direction reversed: misread as positive
LLM Sentiment	-1.00	Strong negative: accurately captures skepticism
LLM Strategic Planning ratio	0.57	Strong signal: far above P75=0.20

Representative Q&A: Retail investors questioned "Why only one price hike in three years? Is management too conservative?", "Is the control-volume-control-price policy a strategic error?", and "What is the company's 5-year product plan?". The LLM classified these as Strategic Planning (ratio 0.57) with sentiment -1.00. The lexicon, seeing words like "plan," "future," and "market," misclassified the discussion as slightly positive (+0.09).

Post-event return: CAR_0_20 = +16.87%. After the price hike announcement, the market quickly interpreted it as an earnings-elasticity catalyst, and retail skepticism converted to relief. Dongwu Securities commented: "The 29% price increase after three years highlights scarcity and noble value, bringing earnings elasticity."

5.2 TCL Technology (000100.SZ) — May 8, 2023

Background: In early May 2023, TCL Technology was at the inflection point of the panel-industry cycle reversal. Q1 semiconductor display revenue of 15.12 billion RMB returned to quarter-over-quarter profit. Panel prices rose at the start of May (65" prices up 31.13% from September 2022 trough). TCL CSOT was showing the world's first 65" 8K flexible printed-OLED foldable TV at SID 2023, and the 35-billion-RMB t9 project (8.6G oxide-semiconductor production line) continued.

Method	Signal Value	Interpretation
Attention z-score	+0.03	Normal levels: traditional methods cannot identify
Lexicon lex_score	+0.57	Strong positive: direction completely reversed
LLM Sentiment	-0.25	Weakly negative: captures skepticism

LLM Strategic Planning ratio	0.50	Strong signal

Representative Q&A: Investors questioned whether panel price increases could persist, the t9 project's 35-billion-RMB investment payback period, and whether persistent solar (TCL Zhonghuan) losses indicated a failed diversification strategy.

Post-event return: CAR_0_20 = +11.53%. Subsequent Q3 earnings validated the strategic skepticism—semiconductor display revenue grew 73% YoY with quarterly profit return, and net income attributable to parent grew 474% YoY.

5.3 CATL (300750.SZ) — February 24, 2022

Background: February 24, 2022 coincided with the outbreak of the Russia-Ukraine war and severe global market turbulence. CATL faced multiple pressures: stock price fell below 500 RMB/share with market cap losing nearly 200 billion RMB in four days. Lithium-carbonate prices were surging. Against this backdrop, the company launched the CTC integrated electric chassis development project in Shanghai (strategic-level commitment).

Method	Signal Value	Interpretation
Attention z-score	+0.19	Slightly above industry baseline
Lexicon lex_score	+0.25	Weakly positive
LLM Sentiment	+0.09	Slightly positive (different from other cases)
LLM Strategic Planning ratio	0.55	Strong signal

Representative Q&A: Investors deeply discussed Kirin Battery/CTP3.0 mass production timing, sodium-ion battery technology routes, overseas plant construction progress, and cost-pass-through capability. This is a "constructive strategic discussion" case rather than a "skepticism wave" case.

Post-event return: CAR_0_20 = +7.97%. After the Russia-Ukraine shock recovered, markets reassessed CATL's long-term electrification trends. This case demonstrates that the Strategic Planning signal is not limited to skepticism-wave contexts—any form of intensive strategic discussion has predictive value.

5.4 Midea Group (000333.SZ) — April 15, 2024

Background: April 2024 was a critical milestone for Midea's globalization strategy: Chairman Fang Hongbo and NIO's Li Bin signed a strategic cooperation agreement on April 10; the 2023 annual shareholders meeting in April addressed investor questions about the Hong Kong IPO; and the company re-submitted its H-share application to the HKEX in April.

Method	Signal Value	Interpretation
Attention z-score	-0.63	Below industry: traditional miss

Lexicon lex_score	+0.71	Strong positive: most severe direction reversal
LLM Sentiment	-0.22	Weakly negative: captures skepticism
LLM Strategic Planning ratio	0.44	Significant signal

Representative Q&A: Retail investors questioned the necessity of HK listing (why fundraise given ample cash), whether H-share discount harms A-share shareholders, and the effectiveness of B-to-B diversification.

Post-event return: CAR_0_20 = +5.93%. This case is particularly valuable: it proves that even in 2024 (the year of overall effect decay), valid case examples exist—suggesting that signals are not failing, but becoming more selective.

5.5 Four-Case Composite Evidence

Method	4-Case Mean	Interpretation
LLM Strategic Planning ratio	0.51	Far above P75=0.20; 4/4 perfect identification
Attention z (traditional)	-0.29	Below industry; traditional methods systematically miss
Lexicon lex_score (traditional)	+0.40	4/4 show positive direction—completely reversed
LLM Sentiment	-0.35	Weakly negative—captures skepticism (but opposite to returns)

This table is the study's most striking empirical evidence—using real cases to demonstrate the two failure modes of traditional methods (false negatives and direction reversal) and the successful identification by LLM methods.

6. Discussion

6.1 Why Does Sentiment Direction Oppose Price Direction?

The Case Studies present a counterintuitive finding—across 4 cases, average LLM sentiment is -0.35 (negative), yet subsequent 20-day CAR average is +10.58% (strongly positive). This is not a contradiction but the study's deepest finding.

Theoretical explanation (retail sentiment as contrarian indicator):

- Retail sentiment lags events: real event occurs → stock price reacts (pre-event drift) → retail perceives price changes and emotional response → retail rushes to Hudongyi to ask → LLM identifies "event-window negative sentiment." By this point, most negative information is priced in, so post-event prices rebound.
- Institutional contrarianism: Classical De Long et al. (1990) noise-trader theory: institutions buy when retail is extremely negative.
- Mean reversion: extreme retail sentiment corresponds to overselling, with subsequent mean reversion bringing positive returns.

This aligns perfectly with Tetlock (2007, JoF)—media pessimism correlates negatively with future returns. This study replicates this classical finding in the A-share, Hudongyi, LLM context.

6.2 Why Are Topic Signals Immune to This Problem?

The key distinction is the fundamental nature of sentiment vs. topic:

- Sentiment: reflects what retail investors feel (panic, worry, excitement)—about the investors themselves, the downstream of events.
- Topic: reflects what retail investors discuss (which aspect of the company)—about the company, the event type label.

When retail concentrate on "Strategic Planning"—regardless of whether sentiment is skeptical (Pien Tze Huang) or constructive (CATL)—the company is at a strategic-importance juncture. The strategic juncture itself (a fundamental event) is what predicts future prices.

Analogy: sentiment = whether the patient is groaning (reflects suffering but tells you nothing about the cause); topic = which body part the patient is complaining about (directly points to the problem).

6.3 Theoretical Contributions

This study contributes on two levels to the quantitative research paradigm:

Critical evidence: Social-media-based "sentiment factors" may be essentially momentum-factor proxies. Many similar studies have failed out-of-sample over the past decade; this study provides a mechanistic explanation—sentiment is a mirror of pre-event price drift, not a predictor.

Constructive direction: Topic signals bypass the sentiment-as-contrarian-indicator trap because they extract event types rather than retail reactions. This provides a clear positioning for the unique value of LLMs in quantitative research—the core value is not "more accurate sentiment scoring" but "extracting topic dimensions."

6.4 Limitations

- Sample scope: Only CSI 300 constituents; generalizability to small caps is limited.
- Time range: Only 3 years (2022-2024); the 2024 decay suggests signal instability.
- LLM consistency: Topic Kappa = 0.553 is only "moderate," which may affect topic-grouping precision.
- Practical feasibility: Sharpe of 0.40 is unlikely to support a standalone strategy after transaction costs.
- No live-trading validation: Only historical backtest economic significance is quantified; out-of-sample performance is not rigorously tested.

6.5 Future Research Directions

- Extend the sample to CSI 500 or the full market to test signal effectiveness on smaller caps.
- Introduce additional topic dimensions (e.g., ESG, geopolitical risk, technology change).
- Integrate LLM topic signals with traditional multi-factor models and quantify alpha contribution.
- Investigate the mechanism of "topic-signal decay": is it market-efficiency improvement or LLM classification baseline drift?

7. Conclusion

This study systematically compares four signal-extraction methods applied to A-share investor Q&A data within an event-study framework. Based on 286 CSI 300 constituents, 1,164 events from 2022-2024, and 17,807 LLM-annotated Q&As, three core conclusions emerge:

First, traditional methods (attention baseline, dictionary-based sentiment) fail completely in a rigorous event-study framework (0 of 7 windows significant), while LLM methods (sentiment + topic) are significantly effective (6 of 7 windows significant)—strong evidence of a qualitative difference between LLM and traditional approaches.

Second, LLM sentiment and topic signals differ fundamentally. The sentiment signal appears strong in univariate grouping (max $|t| = 5.71$), but deeper analysis shows its predictive power comes mainly from mirroring pre-event price drift—after controlling for market state, joint significance disappears ($p = 0.85$). Topic signals remain jointly significant after rigorous controls (topic F-test $p = 0.002$), with "Strategic Planning"-heavy events showing CAR_0_20 exceeding low-topic events by +2.92% ($p = 0.001$).

Third, the mechanism by which topic signals bypass the sentiment-contrarian trap: sentiment reflects retail reactions to events (downstream), while topic reflects event types (the events themselves). What is asked carries more predictive value than the emotion behind the questions.

The theoretical implications are significant: critical evidence for the limits of social-media-based sentiment quantification, and a constructive direction for LLM applications in quantitative research—shifting from "measuring sentiment" to "identifying topics." In a retail-dominated market like A-shares, LLM-extracted topic signals from investor discussions may be a potential alpha source for future multi-factor models.

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Appendix: Data Sources

All data used in this study come from publicly accessible or compliant commercial databases:

- Investor Q&A: Shenzhen Hudongyi (http://irm.cninfo.com.cn); Shanghai E-Interactive (http://sns.sseinfo.com)
- Market and fundamental data: Tushare Pro (https://tushare.pro) — daily stock prices, adjustment factors, index daily data, Shenwan industry classification, announcement data
- LLM Annotation: DeepSeek-V3 API (https://api.deepseek.com)
- Chinese financial sentiment lexicon: Dalian University of Technology Sentiment Dictionary (publicly released)
- Case Study background research: East Money, CNINFO, Sina Finance, ChinaNews, 21st Century Business Herald, and other public news sources

All code, data-processing scripts, and intermediate outputs are archived and available for replication.